

Teorema de Hahn-Banach II

Teorema 1 (de Hahn-Banach II). Sea X un espacio vectorial normado, $M \subset X$ un subespacio y $T: M \rightarrow \mathbb{K}$ lineal y continua

$$\implies \exists F \in \mathcal{L}(X, \mathbb{K}) : \begin{cases} \forall x \in M : F(x) = T(x) \\ \|F\| = \|T\|. \end{cases}$$

Demostración: Como T es continua, por el `teo-carac-continuidad-apl-lineal`/Teorema 1, $\forall x \in M : |T(x)| \leq \|T\| \|x\|$. Tomando $p(x) = \|T\| \|x\|$, tenemos que p es una seminorma en X y $\forall x \in M : |T(x)| \leq p(x)$.

Por tanto, por el `teo-hahn-banach-i`/Teorema 1, existe $\exists F: X \rightarrow \mathbb{K}$ lineal tal que $\forall x \in X : |F(x)| \leq p(x)$ y $F|_M = T$. Entonces, $\|F\| = \sup_{\|x\|=1} |F(x)| \leq \sup_{\|x\|=1} p(x) = \|T\|$.

Por otro lado, $\|F\| = \sup_{\|x\|=1} |F(x)| \geq \sup_{\substack{x \in M \\ \|x\|=1}} |F(x)| = \sup_{\substack{x \in M \\ \|x\|=1}} |T(x)| = \|T\|$. ■

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